PRACTICE QUESTIONS JOINT CONTINUOUS RANDOM VARIABLES

EXERCISE – 3.4

JOINT DENSITY FUNCTIONS

1. A fast-food restaurant operates both a drive through facility and a walk-in facility. On a randomly selected day, let X and Y, respectively, be the proportions of the time that the drive-through and walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x,y)=\begin{cases}\frac{2}{3}(x+2y) & 0\le x\le 1,\ \ 0\le y\le 1\\ 0 & elsewhere\end{cases}$$

(a) Find the marginal density of X.
(b) Find the marginal density of Y.
(c) Find the probability that the drive-through facility is busy less than one-half of the time

$$(Ans: g(x)=\frac{2}{3}(x+1), h(y)=\frac{1}{3}(1+4y), \frac{5}{12})$$

2. Each rear tire on an experimental airplane is supposed to be filled to a pressure of 40 pounds per square inch (psi). Let X denote the actual air pressure for the right tire and Y denote the actual air pressure for the left tire. Suppose that X and Y are random variables with the joint density function

$$f(x,y)=\begin{cases}k(x+2y) & 30\le x\le 50,\ \ 30\le y\le 50\\ 0 & elsewhere\end{cases}$$

(a) Find k.
(b) Find $P(30 \le X \le 40$ and $40 \le Y < 50)$.
(c) Find the probability that both tires are under filled

$$(Ans: 3/3920000, 40/196, 37/196)$$

3. Consider the following joint probability density function of the random variables X and Y:

$$f(x,y)=\begin{cases}\frac{3x-y}{9} & 1<x<3,\ \ 1<y<2\\ 0 & elsewhere\end{cases}$$

(a) Find the marginal density functions of X and Y.
(b) Are X and Y independent?
(c) Find $P(X > 2)$.

$$(Ans: g(x)=\frac{2x-1}{6}, h(y)=\frac{12-2y}{9}, Not\ Independent, 2/3)$$

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4. Two electronic components of a missile system work in harmony for the success of the total system. Let *X* and *Y* denote the life in hours of the two components. The joint density of *X* and *Y* is

$$f(x,y)=\begin{cases} y\,e^{-y(1+x)} & x,y\geq 0 \\ 0 & elsewhere \end{cases}$$

(a) Give the marginal density functions for both random variables.
(b) What is the probability that the lives of both components will exceed 2 hours?
$(Ans: g(x)=\dfrac{1}{(1+x)^2}, h(y)=e^{-y}, \dfrac{1}{3e^6})$

5. The random variables X and Y are jointly distributed as

$$f(x,y)=\begin{cases} 24x^2y(1-x) & 0\leq x,y\leq 1 \\ 0 & elsewhere \end{cases}$$

Find
(a) marginal distributions of X
(b) conditional distribution of Y.
(c) Are X and Y independent?
$(Ans: g(x)=12x^2(1-x), f(y/x)=2y, X\ and\ Y\ are\ independent)$

6. The joint density function of the random variables *X* and *Y* is given by

$$f(x,y)=\begin{cases} x+y & 0<x<1 \quad 0<y<1 \\ 0 & otherwise \end{cases}$$

Find
a. $P(0\leq X\leq 1/2\ and\ 1/2\leq Y\leq 1)$
b. $P(X+Y<1)$
c. Are X and Y independent?
$(Ans: 1/4,\ 1/3,\ Not\ Independent)$

7. The joint density function of the random variables *X* and *Y* is given by

$$f(x,y)=\begin{cases} \dfrac{6}{7}\left(x^2+\dfrac{xy}{2}\right) & 0\leq x\leq 1 \quad 0\leq y\leq 2 \\ 0 & elsewhere \end{cases}$$

Find the following
(a) $P(X>Y)$
(b) $Corr(X,Y)$
$(Ans: 15/56, \quad)$

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8. The joint density function of the random variables *X* and *Y* is given by

$$f(x,y)=\begin{cases} x^2+\frac{1}{3}xy & 0\le x\le 1 \quad 0\le y\le 2 \\ 0 & elsewhere \end{cases}$$

Find

a. $P(X>1/2)$
b. $P(Y<1/2)$
c. $P(Y<X)$
d. $P(X+Y>1)$
e. $P\left(Y<\frac{1}{2}\middle| X<\frac{1}{2}\right)$

$(Ans: 5/6, \quad , 7/24, 65/72, 5/32)$

9. Suppose that in a certain drug the concentration of a particular chemical is a random variable with the following continuous distribution

$$g(x)=(3/8)x^2 \quad for \quad 0\le x\le 2 \ \& \ 0 elsewhere$$

Suppose that the concentrations X and Y of the chemical in two separate batches of the drug are independent random variables each with the same p.d.f . Determine the following

(a) the joint p.d.f of X and Y and verify that it is a valid p.d.f.
(b) $P(X>Y)$
(c) $P(X+Y\le 1)$
(d) $find\ Corr(X,Y)$
$(Ans:)$

10. If X is the amount of money (in hundreds of rupees) that a salesman spends on gasoline during a day and Y is the corresponding amount of money in (in hundred of rupees) for which he is reimbursed, the joint density of these two random variables is given by

$$f(x,y)=\begin{cases} K\frac{20-x}{x} & 10<x<20 \quad \frac{x}{2}<y<x \\ 0 & elsewhere \end{cases}$$

$Find\ Corr(X,Y)$

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